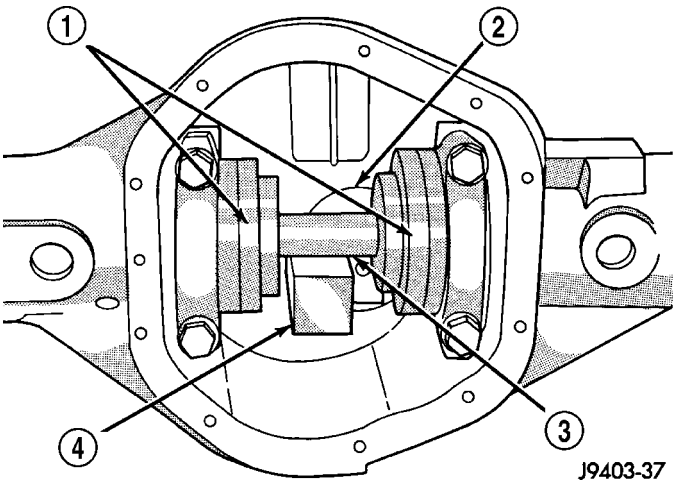


Fig. 8 GAUGE TOOLS IN HOUSING

1 - ARBOR DISC
2 - PINION BLOCK
3 - ARBOR
4 - PINION HEIGHT BLOCK

(5) Assemble Dial Indicator C-3339 into Scooter Block D-115-2 and secure set screw.

(6) Place Scooter Block/Dial Indicator in position in axle housing so dial probe and scooter block are flush against the rearward surface of the pinion height block. Hold scooter block in place and zero the dial indicator face to the pointer. Tighten dial indicator face lock screw.

(7) Slowly slide the Scooter Block/Dial Indicator across the pinion height block over to the arbor bar (Fig. 9). Move the Scooter Block/Dial Indicator to the crest of the arbor bar and record the highest reading.

(8) Select a shim equal to the thickest dial indicator reading plus or minus the drive pinion gear depth variance number etched in the face of the pinion gear (Fig. 5).

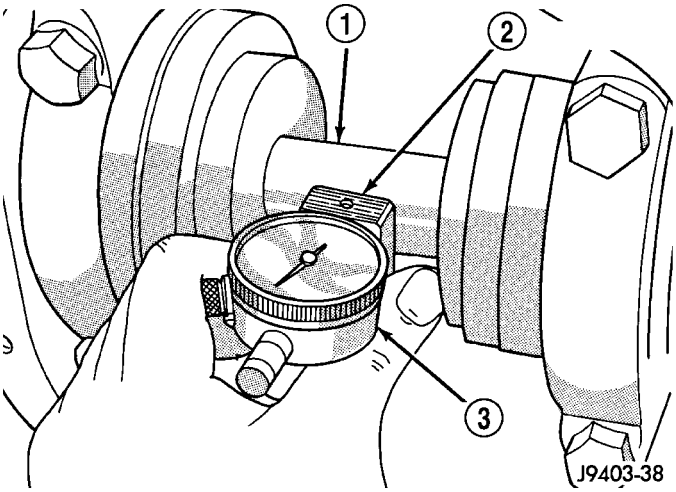


Fig. 9 PINION DEPTH MEASUREMENT

1 - ARBOR
2 - SCOOTER BLOCK
3 - DIAL INDICATOR

(9) Remove the pinion depth gauge components from the axle housing.

PRELOAD SHIM SELECTION

Differential side bearing preload and gear backlash is achieved by selective shims inserted between the bearing cups and the housing. The proper shim thickness can be determined using slip-fit dummy bearings D-345 in place of the differential side bearings and a dial indicator C-3339. Before proceeding with the differential bearing preload and gear backlash measurements, measure the pinion gear depth and prepare the pinion gear for installation. Establishing proper pinion gear depth is essential to establishing gear backlash and tooth contact patterns. After the overall shim thickness to take up differential side play is measured, the pinion gear is installed, and the gear backlash shim thickness is measured. The overall shim thickness is the total of the dial indicator reading and the preload specification added together. The gear backlash measurement determines the thickness of the shim used on the ring gear side of the differential case. Subtract the gear backlash shim thickness from the total overall shim thickness and select that amount for the pinion gear side of the differential (Fig. 10).

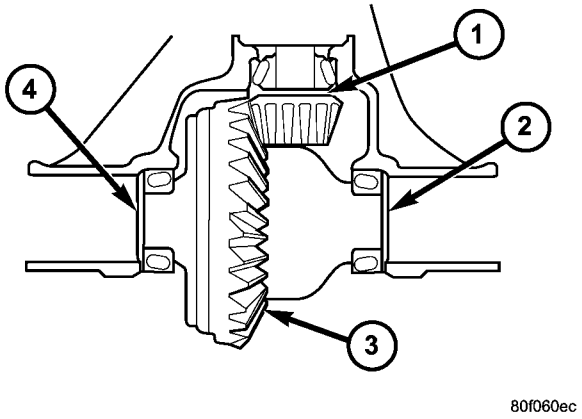


Fig. 10 ADJUSTMENT SHIM LOCATIONS

1 - PINION DEPTH SHIM
2 - DIFFERENTIAL BEARING PRELOAD SHIM
3 - RING GEAR
4 - DIFFERENTIAL BEARING PRELOAD SHIM

NOTE: It is difficult to salvage the differential side bearings during the removal procedure. Install replacement bearings if necessary.

(1) Remove differential side bearings from differential case.

(2) Remove factory installed shims from differential case.

(3) Install ring gear on differential case and tighten bolts to specification.